

CONCEPT 03

Spatial reaction and memory

Targets appear around the real room. The user reaches and touches them with a fingertip, tracked as a collision against the hand joint. A 3D Corsi task lights cubes on room surfaces in sequence and the user repeats the order.

Output: reaction time, movement time, error rate, Corsi span, and reach time split by target eccentricity.

CORE PICK

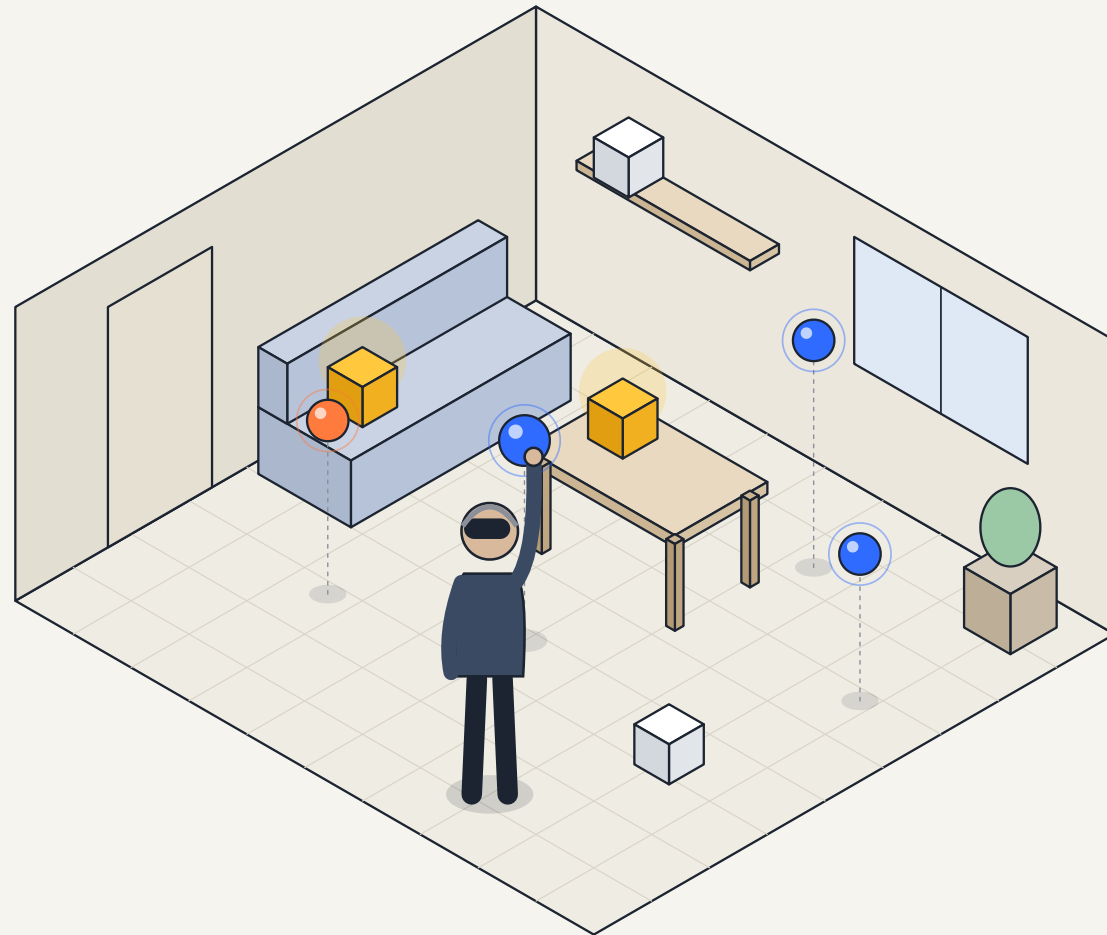
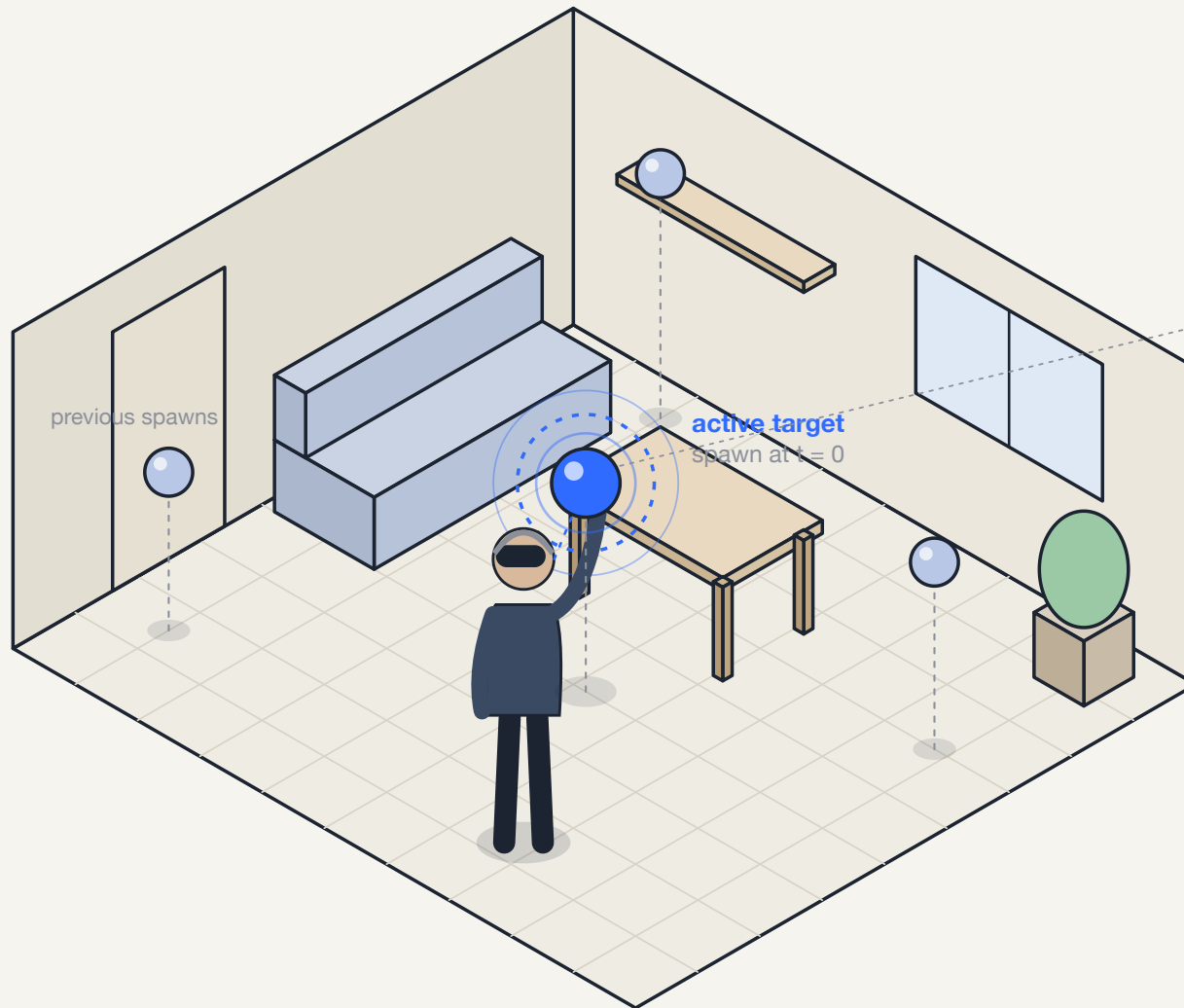


FIG 01

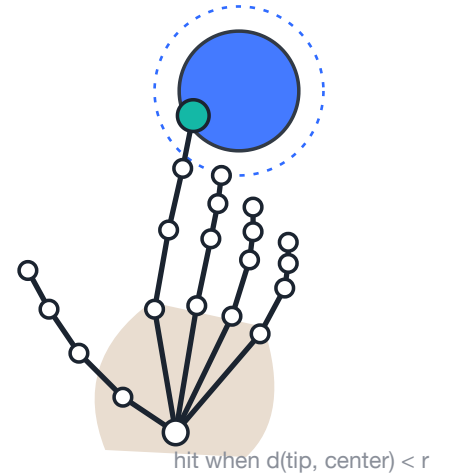
Simple reaction

A target spawns somewhere in the real room. Reach it, touch it with a fingertip.



COLLISION

index tip joint vs target



● active

● earlier trials

dashed lines drop to the floor to show height

TASK

One target at a random point in the room. Trial ends when the index tip joint enters the target radius.

MEASURES

Reaction time to hand start, movement time to contact.

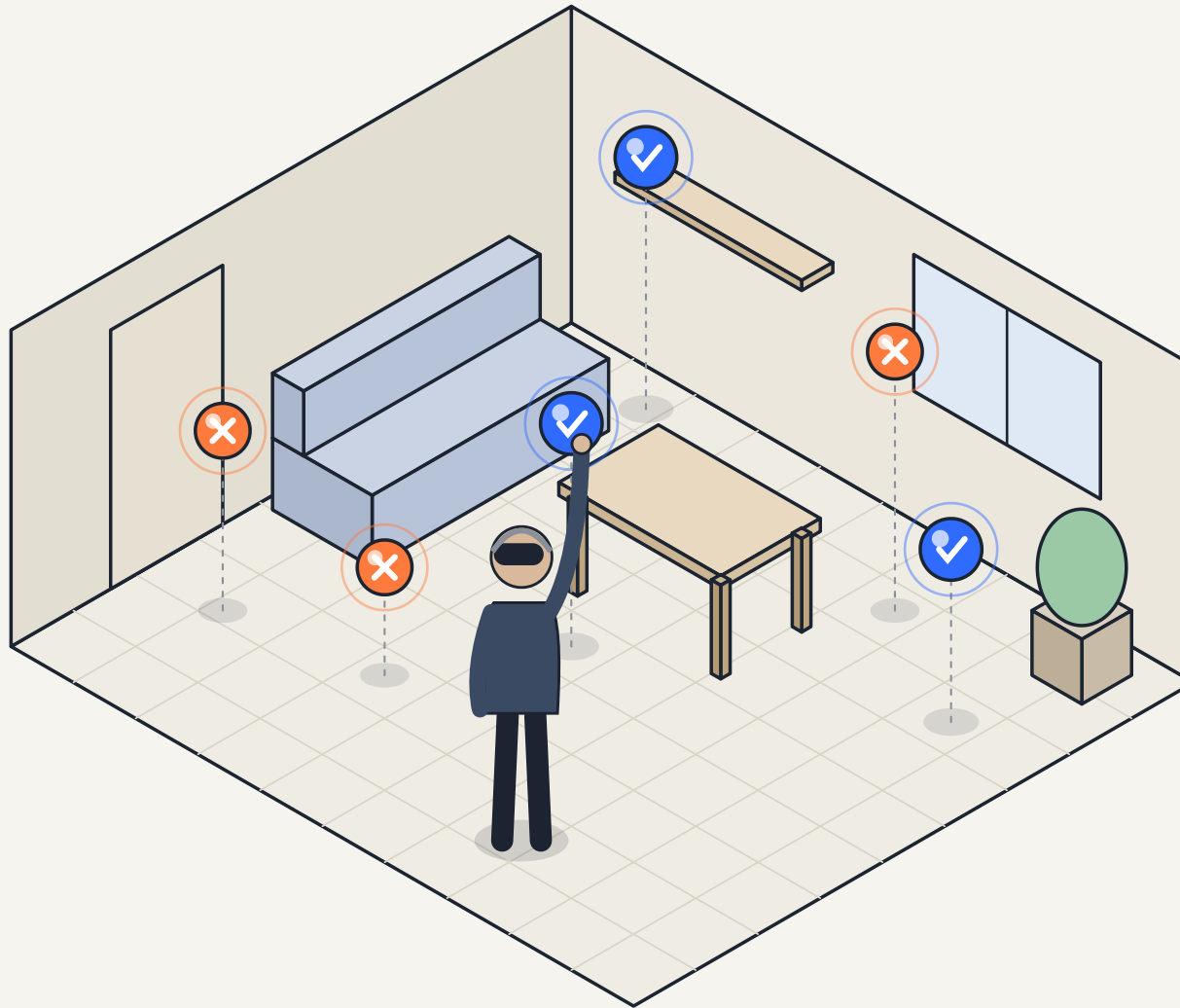
PLACEMENT

Spawn points sampled from the scanned room volume within arm reach.

FIG 02

Choice reaction

Several targets appear at once. Touch only the blue ones.



RULE

touch
counts as a hit

ignore
touching = commission error

missed blue after 2.0 s
= omission error

ratio blue : orange varies

● go ● no-go

TASK

Blue and orange targets appear together. Only blue counts.

ERRORS

Touching orange is a commission error. A blue left untouched past the window is an omission error.

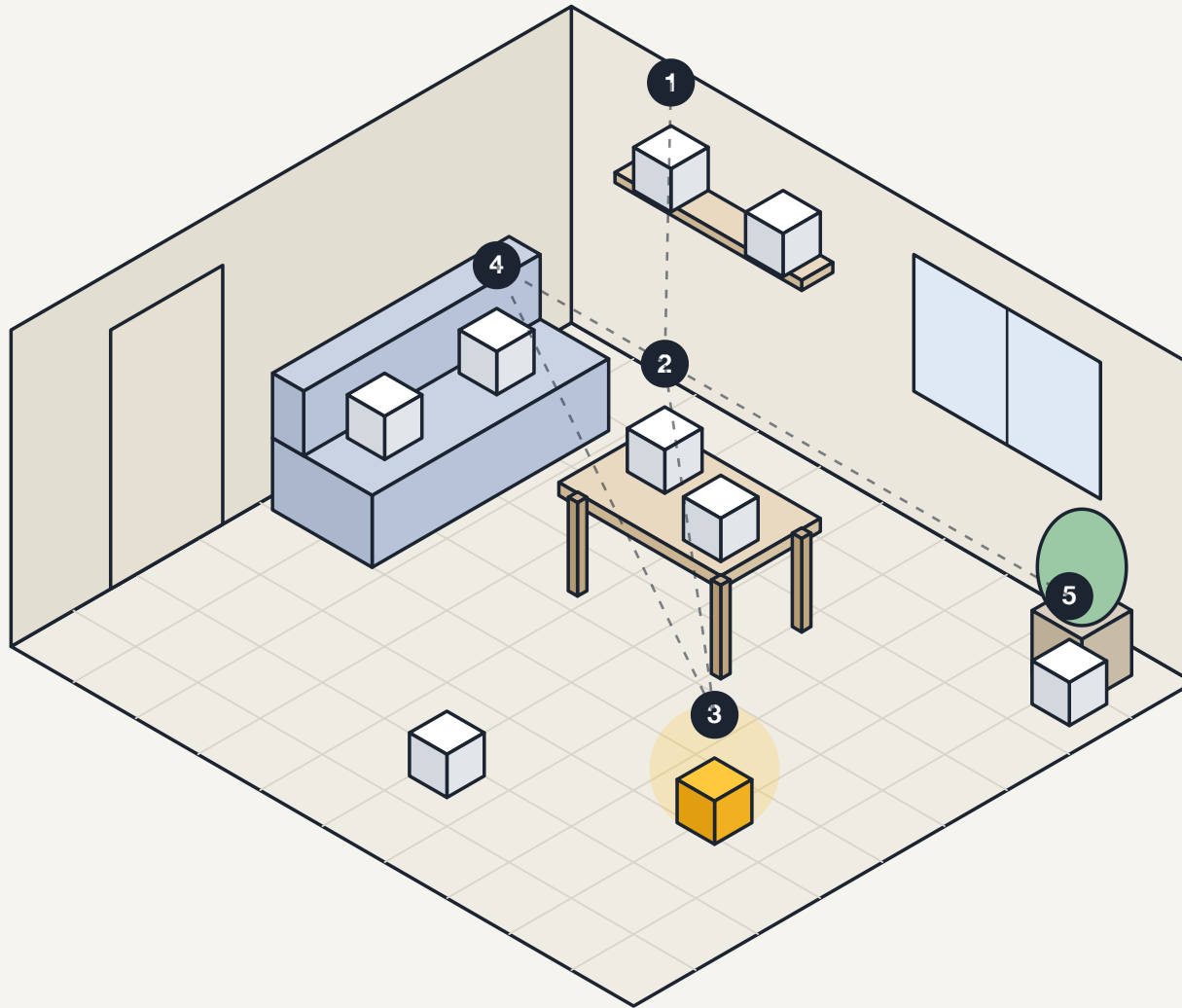
LOAD

Choice adds a decision step, so choice RT minus simple RT isolates decision time.

FIG 03

3D Corsi blocks

Cubes on real surfaces light in sequence. The user repeats the order by touch.



TRIAL

- 1 Show**
cubes light one at a time, 1 s each
- 2 Hold**
short blank, cubes all dim
- 3 Repeat**
user touches cubes in order
- 4 Adapt**
2 correct: span +1. 2 wrong: stop

Corsi span = longest correct run

sequence



step 3 lit now, span length 5

TASK

Cubes anchored to table, shelf, sofa and floor. A sequence lights, then the user touches it back in order.

ADAPTIVE

Span grows by one after two correct trials and stops after two failures at a length.

SCORE

Corsi span is the longest sequence repeated correctly.

FIG 04

Measured features

Every trial yields timing and accuracy. Reach time is split by eccentricity.



FEATURES

Reaction time, movement time, error rate, Corsi span.

ECCENTRICITY

Reach time binned by angle between head forward and target. Targets behind the user need a head turn first.

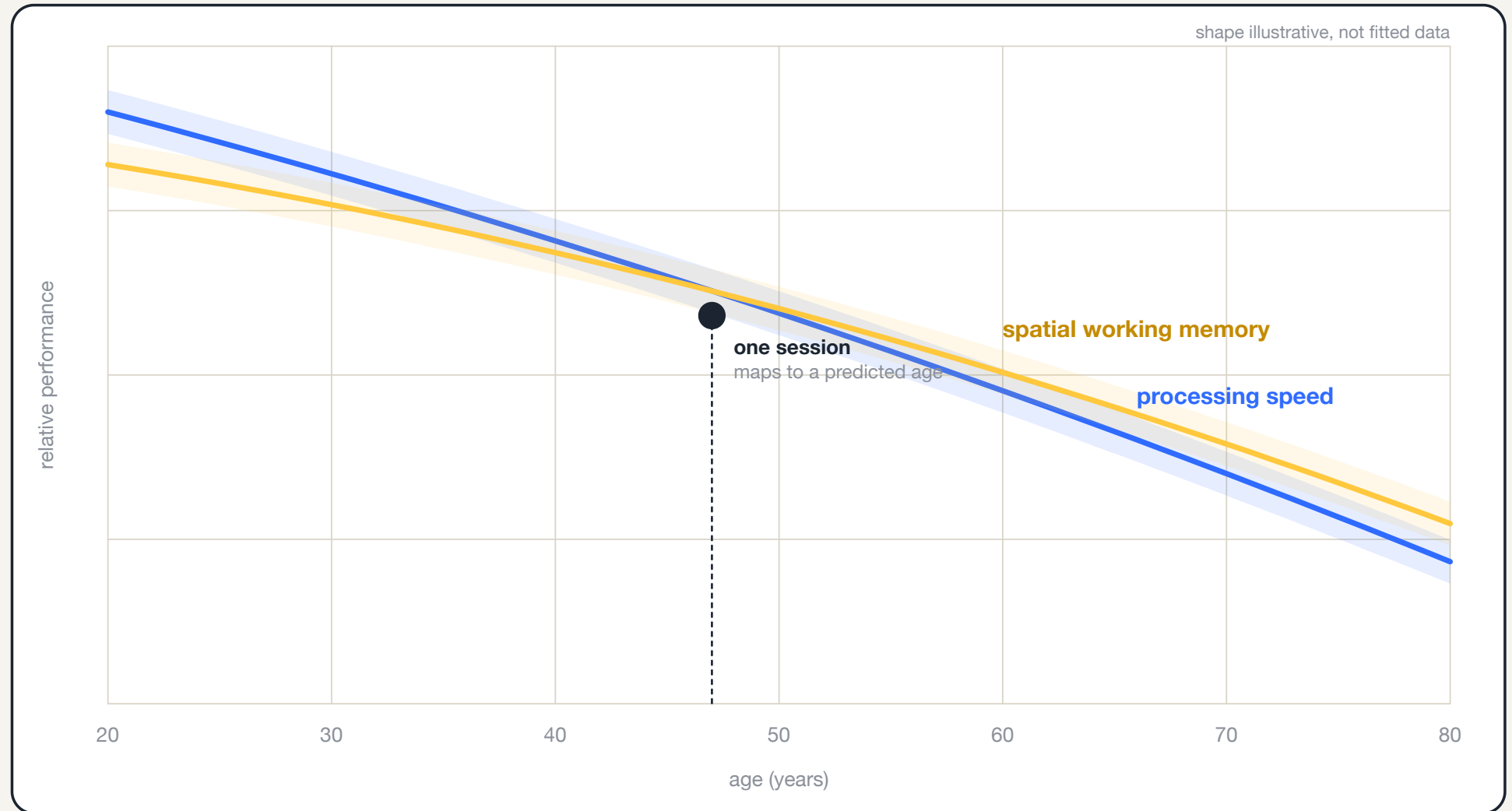
VALUES

Numbers shown are placeholders for layout.

FIG 05

Why it tracks age

Processing speed and spatial working memory fall steadily from early adulthood.



WHY IT AGES

Processing speed and spatial working memory decline steadily from early adulthood, with well documented curves.

USE

A session's features place the user on these curves to estimate a functional age.

NOTE

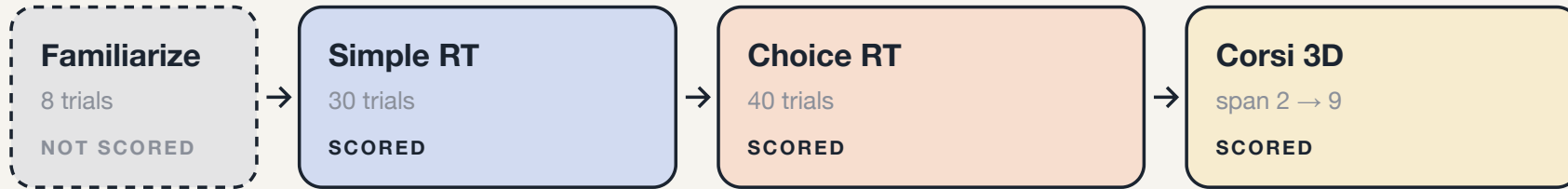
Curve shapes are drawn for illustration.

FIG 06

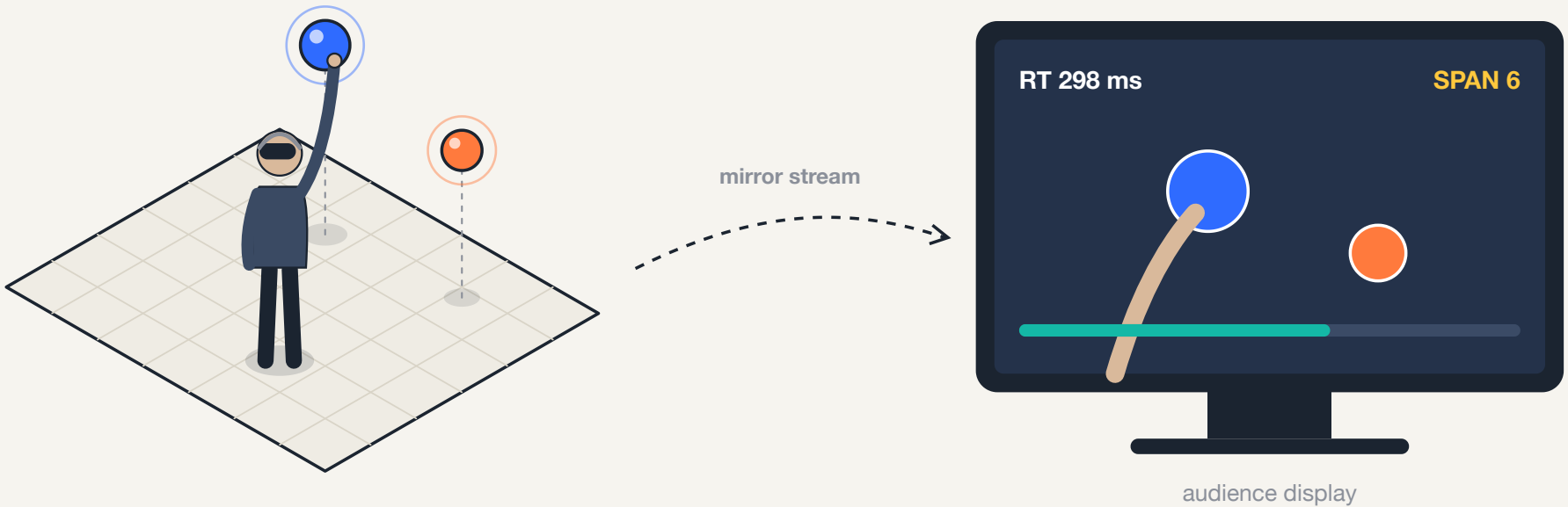
Session and demo view

A fixed warm-up absorbs practice effects. The audience watches a mirrored view.

SESSION, ABOUT 6 MIN



scores start after this line



DEMO

Fast and game-like. The audience follows along on the mirrored view.

RISK

Practice effects inflate scores on repeat sessions.

MITIGATION

A fixed short familiarization round runs first. Only trials after it are scored.